Industrial Security (Master)

Exercise Sheet – Section 02

PLC Ladder Logic Reverse Engineering

Exercise 2.1 – Project vs Runtime

On a PLC you own (OpenPLC, or a spare hardware unit):

1. Write and download a small program (motor seal-in).
2. Save the project file and compute its SHA-256.
3. Upload the program blocks back from the PLC; compute SHA-256.
4. Modify one rung directly on the runtime (without changing the project file). Re-upload. Compare hashes. Document what changed.

Exercise 2.2 – Snap7 Inspection

Using `python-snap7`:

1. List all blocks on the PLC.
2. Upload OB1 and one DB; dump the raw MC7 bytes.
3. Identify the block header, local data size, and the code section by hand.

Exercise 2.3 – Logic-Bomb Hunt

Given a project file from the instructor:

1. Identify all unused inputs and all time-/counter-based branches.
2. For each suspicious branch, infer the condition under which it fires.
3. Write a 1-page memo describing the hidden behaviour and its consequences.

Exercise 2.4 – Defence Pitch

Pick one of the defensive controls from the slides (signed downloads, key-switch, checksum monitoring, version-controlled projects). In 1 page:

- Sketch how to implement it on Siemens S7-1500 *and* Rockwell ControlLogix.
- Identify the trade-off (operational cost, vendor support, downtime).
- Recommend a 12-month rollout plan.